ELECTRODYNAMICS II · SEMESTER 5

# Fields in matter

Module 1 · Prepare

Start with the idea. Learn the mathematics. Work the questions. Check the result.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

## How to use this book

P means prerequisite, M means mathematics, T means topic and Q means the numbered practice question. Topic codes restart in each module. Read P1 and M1 first, then follow the topics in order.

**REMIND ME**

A fact from school that we will use again.

**MATHS TOOL**

The meaning of a symbol and the operation it asks you to do.

**THE IDEA**

A physical explanation before the equation.

**RESULT**

The statement to carry into an exam.

**EXAM TRAP**

An assumption, sign or unit that changes the answer.

**CHECK YOURSELF**

Cover the answer and try. Then explain why your answer makes sense.

# One rule for this module

> **KEEP THIS IN MIND**
>
> Matter responds through tiny electric dipoles and magnetic dipoles. Account for that response before applying the field laws.

## The whole module in five sentences

1. An electric field slightly separates charges or turns existing dipoles.
2. Their average dipole moment per volume is polarization P, whose edges and nonuniformity act like bound charge.
3. D counts free charge while E describes the actual electric force.
4. Microscopic magnetic moments make magnetization M, equivalent to bound currents.
5. H separates free-current excitation from material response; ferromagnets also remember their history.

## Syllabus route

Units 1–5 · 12 hours · 16 theory marks. Griffiths, 5th ed.: §§4.1.1–4.1.4, 4.2.1–4.2.2, 4.3.1, 4.3.3, 4.4.1; §§6.1.1–6.1.4, 6.2.1–6.2.3, 6.3.1, 6.4.1–6.4.2.

Definitions and derivations are independently explained from the named textbook sections. Worked numerical values are original practice unless a past-paper tag is printed.

# Topic route

Unit 1 → T1–T3; unit 2 → T4; unit 3 → T5–T7; unit 4 → T8–T9; unit 5 → T10–T13.

T1 · Dielectrics and induced dipoles
T2 · Permanent dipoles: torque and force
T3 · Polarization as a density
T4 · Bound charge and the field of polarization
T5 · Displacement and free charge
T6 · Electric boundary conditions
T7 · Linear dielectrics and capacitors
T8 · Magnetic response of atoms
T9 · Magnetization and magnetic torque
T10 · Bound currents and magnetized objects
T11 · The auxiliary field H
T12 · Magnetic susceptibility and permeability
T13 · Ferromagnetism and memory

# P1 School physics bridge

## Charge, field and potential

Charge $q$ is measured in coulombs (C). Current $I = \Delta q/\Delta t$ is charge passing a point per second, measured in amperes (A). An electric field is force per positive test charge: $\mathbf{E} = \mathbf{F}/q$, in N/C or V/m. A negative charge feels force opposite to E.

Potential $V$ is potential energy per charge, in volts. A uniform field through distance $d$ gives a potential drop of magnitude $Ed$. A capacitor stores separated charge: $Q = CV$, with capacitance C in farads (F). Its parallel-plate value in vacuum is $C_0 = \epsilon_0 A/d$, neglecting edge effects.

## Dipoles and torque

Two charges $-q, +q$ separated by vector $\mathbf{d}$ have moment $\mathbf{p} = q\mathbf{d}$; d points from minus to plus. Torque turns an object: force × perpendicular distance. Equal opposite forces can give zero net force and a nonzero torque.

## Magnetic quantities

A charge moving with velocity v experiences $\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}$. B is measured in tesla (T). A small planar loop with current I and area A has magnetic moment $\mathbf{m} = IA\hat{\mathbf{n}}$. Curl your right fingers with the current: your thumb gives n and m. This is a loop model, not two isolated magnetic charges.

### Constants and prefixes

Use $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 \simeq 4\pi \times 10^{-7}$ H/m for these exercises. Micro ($\mu$) means $10^{-6}$, nano means $10^{-9}$, milli means $10^{-3}$. Convert centimetres to metres before squaring an area or cubing a volume.

## M1 Mathematics from the beginning

### Vectors: projection and turning

Write $\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$. A hat means a unit vector: it gives direction, with length one. A dot product selects a projection: $\mathbf{A} \cdot \hat{\mathbf{n}} = A\cos\theta$. Thus polarization normal to a face contributes all of P; polarization parallel to it contributes zero.

A cross product has magnitude $AB\sin\theta$ and points perpendicular to A and B. The order matters: $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$, while reversing the order changes the sign.

### A derivative measures a local slope

If $P_x = ax$, increasing x by $\Delta x$ increases $P_x$ by $a\Delta x$. Therefore $dP_x/dx = a$. A partial derivative $\partial/\partial x$ changes x while holding y and z fixed. The gradient $\nabla V$ collects all three slopes. Electric field points downhill: $\mathbf{E} = -\nabla V$.

### Divergence: net outward flow per volume

For a tiny box, more vector field leaving than entering means positive divergence. In Cartesian coordinates, $\nabla \cdot \mathbf{P} = \partial_x P_x + \partial_y P_y + \partial_z P_z$. If $\mathbf{P} =$

$a(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}})$, each derivative is a, so the divergence is 3a.

### Integrals: add tiny pieces

$\int \rho \, d\tau$ adds charge density times tiny volume; for constant density it is just density × volume. $\oint \mathbf{D} \cdot d\mathbf{a}$ adds outward normal flux over a closed surface. The circle on the integral means closed. The divergence theorem says this total surface flux equals $\int (\nabla \cdot \mathbf{D}) d\tau$.

### Curl: local circulation

A field can circulate even when its divergence is zero. Curl measures this circulation per area. For $\mathbf{M} = M_z(x)\hat{\mathbf{z}}$, $\nabla \times \mathbf{M} = -[dM_z/dx]\hat{\mathbf{y}}$. Stokes' theorem turns surface-integrated curl into the circulation $\oint \mathbf{M} \cdot d\mathbf{l}$ around its edge. Curl your fingers around the edge; your thumb fixes its area normal.

### Example of symmetry

If a field has constant magnitude over a spherical surface and is radial, $\oint \mathbf{D} \cdot d\mathbf{a} = D(4\pi r^2)$. Only then can D be taken outside. A field law alone does not guarantee this symmetry.

## T1 Dielectrics and induced dipoles

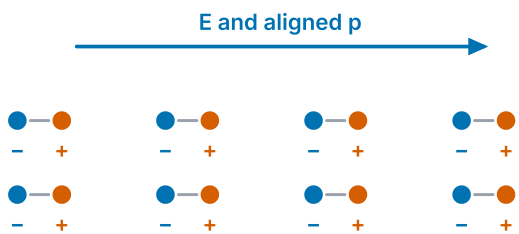
> Purpose: A dielectric can respond to an electric field without allowing charges to travel through the whole sample.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

A dielectric can respond to an electric field without allowing charges to travel through the whole sample.




**Equal charges; separated centres**

Dipole arrows point from minus to plus. Alignment creates polarization without net charge.

1. A nonpolar atom begins with coincident charge centres. The field shifts the positive centre with E and the negative centre against E.
   Why: Opposite charges feel opposite forces.

2. The displacement stops when the internal restoring force balances the external force.
   Why: A stable atom is bound.

3. The induced moment is proportional to the local field for small displacements.
   Why: A linear approximation is valid near equilibrium.

**RESULT**

$\mathbf{p} = \alpha \mathbf{E}_{\text{local}}$ in the weak-field isotropic model.

**EXAM TRAP**

Do not confuse one atom's polarizability $\alpha$ with a bulk susceptibility $\chi_e$.

**CHECK YOURSELF**

**Does a neutral atom need a net charge to become a dipole?**
Answer: No. Separation of its positive and negative charge centres is sufficient.

This topic supports Q1, Q2, Q23.

## Work it through · Q1

Skill: Insulator response. Model · syllabus-based

**How can an electrically neutral dielectric respond to a static electric field?**

Its positive and negative charge centres shift slightly relative to one another, or existing molecular dipoles turn. Charges remain bound rather than crossing the whole material.

**CHECK THE ENDPOINT**

Neutrality does not prevent polarization.

## Work it through · Q2

Skill: Atomic polarizability. **Legacy PYQ N25 Q5 · expanded**

**Define atomic polarizability and give its SI unit.**

For a weak local field, $\mathbf{p} = \alpha \mathbf{E}_{\text{local}}$. Thus $[\alpha] = (\mathrm{C\,m})/(\mathrm{V/m}) = \mathrm{C\,m^2/V}$. The atom is isotropic in this scalar linear model; α describes an individual atom rather than the whole sample.

**CHECK THE ENDPOINT**

Polarizability is induced dipole moment per unit local field.

## Work it through · Q23

Skill: Classical atomic model. Model · syllabus-based

**Model an atom as a point nucleus +q in a uniform spherical cloud −q of radius a. Derive its polarizability for a small nucleus displacement d.**

1. The cloud density is $-3q/(4\pi a^3)$ and its interior field at displacement d is $-qd/(4\pi\epsilon_0 a^3)$.
   Why: Gauss's law for a uniform sphere gives a field proportional to radius.
2. At equilibrium $E = qd/(4\pi\epsilon_0 a^3)$.
   Why: The applied and restoring forces cancel.
3. Use $p = qd$ to obtain $p = 4\pi\epsilon_0 a^3 E$.
   Why: Dipole moment is charge times separation.

**CHECK THE ENDPOINT**

$\alpha = 4\pi\epsilon_0 a^3$ in this classical model.

# T2 Permanent dipoles: torque and force

Purpose: A uniform field turns a permanent dipole; a nonuniform field can also pull it.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

A uniform field turns a permanent dipole; a nonuniform field can also pull it.

1. The two forces are $+q\mathbf{E}$ and $-q\mathbf{E}$. In a uniform field they cancel as forces.
   Why: Both charges see the same E.
2. Their moments add to $pE\sin\theta$.
   Why: The forces act at different positions.
3. If E changes across the dipole, subtract the two local fields and keep the first spatial derivative.
   Why: The dipole is small compared with the scale of variation.

**RESULT**

$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$, $U = -\mathbf{p} \cdot \mathbf{E}$; for a fixed dipole, $\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}$.

**EXAM TRAP**

For an induced dipole, substituting a field-dependent p directly into $U = -pE$ misses the work of inducing it.

**CHECK YOURSELF**

**Which orientation has minimum permanent-dipole energy?**

Answer: Parallel: $U = -pE$.

This topic supports Q3, Q4, Q24.

## Work it through · Q3

Skill: Dipole torque. Model · syllabus-based

**A permanent dipole has moment $2 \times 10^{-29}$ C m in a field $3 \times 10^{5}$ V/m at 30°. Find the torque magnitude.**

$\tau = pE \sin\theta = (2 \times 10^{-29})(3 \times 10^{5})(0.5) = 3 \times 10^{-24}$ N m. The torque turns p toward E.

**CHECK THE ENDPOINT**

$\tau = 3 \times 10^{-24}$ N m.

## Work it through · Q4

Skill: Stable orientation. Model · syllabus-based

**Which orientation of a permanent dipole in a uniform field is stable? Use energy.**

$U = -pE\cos\theta$ is minimum at $\theta = 0$. A small angular displacement produces a restoring torque. The antiparallel orientation is an energy maximum.

**CHECK THE ENDPOINT**

Parallel alignment is stable.

### Work it through · Q24

Skill: Nonuniform dipole force. Model · syllabus-based

**A fixed dipole $\mathbf{p} = p\hat{\mathbf{x}}$ is in $\mathbf{E} = kx^2\hat{\mathbf{x}}$. Find its force at x. Is there a torque?**

1. The positive and negative ends sample different fields. To first order $F_x = p\,dE_x/dx$.
   Why: Subtract their two electric forces.
2. Since $d(kx^2)/dx = 2kx$, $\mathbf{F} = 2pkx\hat{\mathbf{x}}$.
   Why: p is fixed, so only the field is differentiated.
3. The torque is zero because p and E are collinear.
   Why: Their cross product vanishes.

**CHECK THE ENDPOINT**

$\mathbf{F} = 2pkx\hat{\mathbf{x}}$; $\boldsymbol{\tau} = 0$.

## T3 Polarization as a density

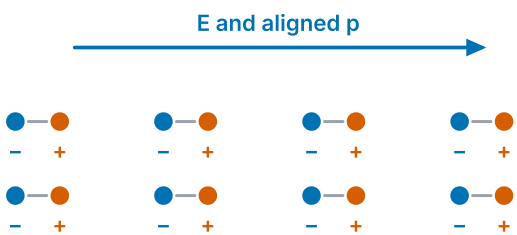
> Purpose: Polarization is the average electric dipole moment contained in a unit volume.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

Polarization is the average electric dipole moment contained in a unit volume.




**Equal charges; separated centres**

Dipole arrows point from minus to plus. Alignment creates polarization without net charge.

1. Add the dipole vectors in a small macroscopic volume.
   Why: Directions matter as well as the number of molecules.
2. Divide by that volume.
   Why: This converts a collection of moments into a local material field.
3. Random directions can give P=0 even when every molecule has a permanent moment.
   Why: Opposite vector contributions cancel.

**RESULT**

$\mathbf{P}$ = dipole moment/volume; units C/m².

**EXAM TRAP**

P is not total charge density and is not the same as the electric field.

**CHECK YOURSELF**

**If number density doubles at fixed aligned moment p, what happens to P?**
Answer: It doubles: $P = np$.

This topic supports Q5, Q6.

## Work it through · Q5

Skill: Polarization definition. Model · syllabus-based

**Define polarization and distinguish it from dipole moment.**

P is electric dipole moment per unit volume (C/m²). The moment p of one dipole has units C m. P is a bulk average and can vanish through random orientation.

**CHECK THE ENDPOINT**

$\mathbf{P} = \Delta\mathbf{p}/\Delta\tau$.

## Work it through · Q6

Skill: Number density conversion. Model · syllabus-based

**A material has $5 \times 10^{27}$ aligned dipoles per m³, each of moment $2 \times 10^{-30}$ C m. Find P.**

Multiply moment per dipole by dipoles per volume: $P = np = 10^{-2}$ C/m², directed along the common dipole direction.

**CHECK THE ENDPOINT**

$P = 0.010$ C/m².

## T4 Bound charge and the field of polarization

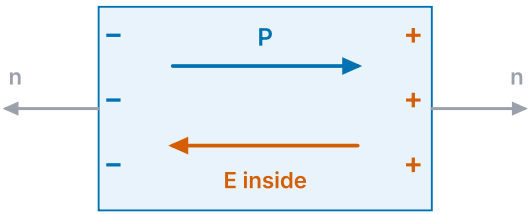
> Purpose: A collection of dipoles can be replaced, for field calculations, by equivalent surface and volume charges.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

A collection of dipoles can be replaced, for field calculations, by equivalent surface and volume charges.




**Uniform slab: E opposes P**

For an infinite uniformly polarized slab with no free charge, bound faces make a depolarizing field.

1. Neighbouring positive and negative ends cancel in the bulk when P is uniform.
   Why: Only unmatched ends remain at faces.
2. Project P onto the outward face normal to find $\sigma_b$.
   Why: Only dipoles crossing that face contribute.
3. For varying P, an imbalance also remains within each tiny volume: $\rho_b = -\nabla \cdot \mathbf{P}$.
   Why: Outward-growing dipole moment leaves negative bound charge locally.
4. Compute V from these equivalent charges, then $\mathbf{E} = -\nabla V$.
   Why: They produce the same external and macroscopic internal field as the dipoles.

**RESULT**

$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$; $\rho_b = -\nabla \cdot \mathbf{P}$.

**EXAM TRAP**

The surface normal must point out of the polarized object; reversing it reverses $\sigma_b$.

**CHECK YOURSELF**

**Uniform P points right in a slab. Which face is positive?**

Answer: The right face, where the outward normal is parallel to P.

This topic supports Q7, Q8, Q25, Q26, Q37.

## Work it through · Q7

Skill: Surface bound charge projection. Model · syllabus-based

**A face normal makes 60° with uniform $\mathbf{P}$, whose magnitude is 0.04 C/m². Find the bound surface charge density.**

$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P\cos 60^\circ = 0.020$ C/m². This sign assumes the stated outward normal.

**CHECK THE ENDPOINT**

$\sigma_b = +0.020$ C/m².

## Work it through · Q8

Skill: Physical interpretation of bound charge.

**Legacy PYQ N25 Q10 · expanded**

**What is the physical interpretation of bound charges? Include the case of uniform polarization.**

Polarization separates positive and negative charge within atoms or molecules, leaving unmatched dipole ends at boundaries and wherever the dipole density varies. These charges remain associated with the material, unlike mobile free charge. For uniform P, $\rho_b = 0$ inside but $\sigma_b = P \cdot \hat{n}$ may remain at its surface.

**CHECK THE ENDPOINT**

Bound charges are the macroscopic result of displaced or aligned microscopic dipoles.

## Work it through · Q25

Skill: Bound charge conservation. Model · syllabus-based

**A sphere of radius R has $\mathbf{P} = a\mathbf{r}$ inside and zero outside. Find its bound charge densities and total bound charge.**

1. $\nabla \cdot \mathbf{r} = 3$, hence $\rho_b = -3a$.
   Why: Each Cartesian coordinate contributes one derivative.

2. At $r = R$, $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{r}} = aR$.
   Why: The sphere's normal is radial.

3. $Q_{b,v} = (-3a)(4\pi R^3/3) = -4\pi a R^3$ and $Q_{b,s} = aR(4\pi R^2) = 4\pi a R^3$.
   Why: Integrate the constant densities over volume and area.

**CHECK THE ENDPOINT**

$\rho_b = -3a$, $\sigma_b = aR$; total bound charge is zero.

## Work it through · Q26

Skill: Depolarizing field of a slab. Model · syllabus-based

**An infinite slab is uniformly polarized perpendicular to its faces, with no free charge. Find E inside and outside.**

1. The faces carry $-P$ and $+P$.
   Why: Use outward normals in $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$.
2. Each infinite sheet produces field magnitude $P/(2\epsilon_0)$.
   Why: Apply Gauss's law to one sheet.
3. Between the sheets the fields add from positive to negative, opposing P; outside they cancel.
   Why: Superpose the two sheet fields.

**CHECK THE ENDPOINT**

Inside $\mathbf{E} = -\mathbf{P}/\epsilon_0$; outside E=0.

## Work it through · Q37

Skill: Derivation of equivalent bound charges.

**Legacy PYQ N25 Q17 · adapted marks**

**Starting from dipole potential, derive the surface and volume bound-charge representation of a polarized object and explain its physical meaning.**

1. Let $\mathcal{R} = |\mathbf{r} - \mathbf{r}'|$, where r is the observation point and r′ labels source points. A volume element has dipole moment $\mathbf{P}(\mathbf{r}')d\tau'$.
   Why: Polarization is moment density.

2. Its potential sums to $V = (4\pi\epsilon_0)^{-1} \int \mathbf{P} \cdot \nabla'(1/\mathcal{R})\, d\tau'$.
   Why: The source-coordinate gradient equals $\hat{\mathcal{R}}/\mathcal{R}^2$.

3. Apply the product rule: $\mathbf{P} \cdot \nabla'(1/\mathcal{R}) = \nabla' \cdot (\mathbf{P}/\mathcal{R}) - (\nabla' \cdot \mathbf{P})/\mathcal{R}$.
   Why: This moves the derivative from the distance factor to the polarization.

4. The divergence theorem converts the first volume integral to a surface integral.
   Why: It equates outward flux and integrated divergence.

5. Thus $V = (4\pi\epsilon_0)^{-1}[\oint \sigma_b\, da'/\mathcal{R} + \int \rho_b\, d\tau'/\mathcal{R}]$.
   Why: These are exactly the potential formulas for surface and volume charge.

6. Identify $\sigma_b = P \cdot \hat{n}$ and $\rho_b = -\nabla \cdot P$. Uniform interior P cancels there, but unmatched dipole ends remain on the surface.
   Why: This explains both the signs and the physical origin.

**CHECK THE ENDPOINT**

The polarized object has the same macroscopic field as its equivalent bound surface and volume charges.

## T5 Displacement and free charge

> Purpose: D packages the material polarization so that its flux counts only free charge.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

D packages the material polarization so that its flux counts only free charge.

1. Gauss's law for E counts total charge: $\epsilon_0 \nabla \cdot \mathbf{E} = \rho_f + \rho_b$.
   Why: Both bound and free charges make E.
2. Insert $\rho_b = -\nabla \cdot \mathbf{P}$ and move that term to the left.
   Why: This groups the unknown material response with E.
3. Define the grouped field as D.
   Why: Its divergence then has only the free source.

**RESULT**

$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$; $\nabla \cdot \mathbf{D} = \rho_f$; $\oint \mathbf{D} \cdot d\mathbf{a} = Q_f$.

**EXAM TRAP**

D is not always determined by free charges alone: boundary conditions and material geometry also matter.

**CHECK YOURSELF**

**If a Gaussian surface encloses no free charge, must D vanish everywhere?**
Answer: No. Its net flux is zero; the local vector can be nonzero.

This topic supports Q9, Q10, Q27.

## Work it through · Q9

Skill: Definition of D. Model · syllabus-based

**Define electric displacement and state its SI unit.**

$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$. Both terms have units C/m². Its closed-surface flux equals enclosed free charge.

**CHECK THE ENDPOINT**

D is measured in C/m².

## Work it through · Q10

Skill: Free versus bound source. Model · syllabus-based

**Which charge appears in Gauss's law for D, and which appears in Gauss's law for E?**

D flux counts $Q_f$. The electric field law is $\epsilon_0 \oint \mathbf{E} \cdot d\mathbf{a} = Q_f + Q_b$. Bound charge is still physically present.

**CHECK THE ENDPOINT**

D counts free charge; E counts total charge.

## Work it through · Q27

Skill: Line free charge in a dielectric.

**Legacy PYQ N25 Q19 · expanded**

**A long straight wire with uniform free line charge λ is surrounded by rubber insulation out to radius a. Find the electric displacement, and state E if the rubber has permittivity ε and the exterior is vacuum.**

1. Assume an infinitely long wire and cylindrical symmetry. A coaxial Gaussian cylinder of radius s and length ℓ has constant radial D on its curved face.
   Why: Translation and rotation symmetry remove axial and angular variation.

2. The cap flux is zero, and curved flux is $D(2\pi s\ell)$.
   Why: D is radial.

3. Enclosed free charge is $\lambda\ell$, so $D = \lambda/(2\pi s)$.
   Why: Gauss's law for D counts free charge only.

4. Thus D has the same expression in rubber and outside. In rubber, $E = D/\epsilon$; outside, $E = D/\epsilon_0$.
   Why: The constitutive relation changes at s=a even though free-charge flux does not.

**CHECK THE ENDPOINT**

$\mathbf{D} = \lambda\hat{s}/(2\pi s)$ outside the wire; E changes with the local permittivity.

# T6 Electric boundary conditions

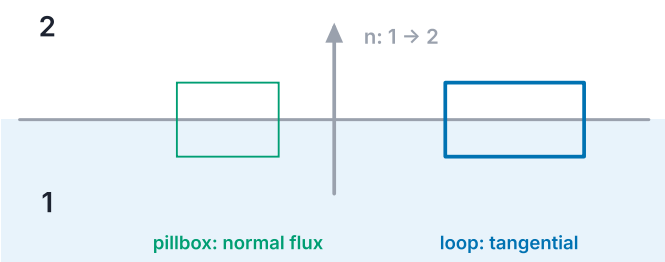
> Purpose: A thin pillbox and a tiny loop tell us which field components can jump at a surface.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

A thin pillbox and a tiny loop tell us which field components can jump at a surface.




A pillbox compares normal fields. A shrinking loop compares tangential fields. Keep the same normal direction in every jump.

1. Shrink a pillbox straddling the interface: the side flux vanishes and the two caps give the normal jump.
   Why: Gauss's law encloses the free surface charge.

2. Shrink a rectangular loop across the interface.
   Why: Electrostatic circulation is zero.

3. Only the two tangential sides survive, so tangential E is continuous.
   Why: The connecting short sides vanish with loop height.

**RESULT**

$D_{2n} - D_{1n} = \sigma_f$; $\mathbf{E}_{2t} = \mathbf{E}_{1t}$ in electrostatics. Normal n points from medium 1 to 2.

**EXAM TRAP**

Normal E need not be continuous. With no free surface charge, normal D is continuous.

**CHECK YOURSELF**

**At an interface with no free charge and $\epsilon_2 = 2\epsilon_1$, how do normal fields compare?**
Answer: $E_{2n} = E_{1n}/2$.

This topic supports Q11, Q12, Q28, Q38.

## Work it through · Q11

Skill: Normal electric discontinuity. Model · syllabus-based

**State the boundary condition for the normal component of D, with your normal convention.**
Let n point from medium 1 to 2. Then $(\mathbf{D}_2 - \mathbf{D}_1) \cdot \hat{\mathbf{n}} = \sigma_f$. This follows from a thin Gaussian pillbox.

**CHECK THE ENDPOINT**

$D_{2n} - D_{1n} = \sigma_f$.

## Work it through · Q12

Skill: Tangential electric continuity. Model · syllabus-based

**Why is tangential E continuous across an electrostatic interface?**

The electrostatic circulation $\oint \mathbf{E} \cdot d\mathbf{l}$ is zero. For a shrinking rectangular loop crossing the surface only its tangential sides survive, so $E_{2t} = E_{1t}$.

**CHECK THE ENDPOINT**

Tangential E has no electrostatic jump.

## Work it through · Q28

Skill: Refraction of electrostatic field lines. Model · syllabus-based

**At an uncharged interface between isotropic dielectrics, derive the relation between field angles $\theta_1, \theta_2$ measured from the normal.**

1. Tangential continuity gives $E_1 \sin\theta_1 = E_2 \sin\theta_2$.
   Why: Electrostatic circulation vanishes.
2. Normal D continuity gives $\epsilon_1 E_1 \cos\theta_1 = \epsilon_2 E_2 \cos\theta_2$.
   Why: There is no free surface charge.
3. Divide the first equation by the second.
   Why: Eliminate the field magnitudes.

**CHECK THE ENDPOINT**

$\tan\theta_2/\tan\theta_1 = \epsilon_2/\epsilon_1$ where the ratio is defined.

## Work it through · Q38

Skill: Dielectric boundary-value method. Model · syllabus-based

**Develop a systematic method for solving a planar dielectric interface problem with free surface charge. Include the field laws and a normal-field example.**

1. Choose normal n from medium 1 into medium 2, and resolve every field into normal and tangential components.
   Why: Sign conventions prevent ambiguous jumps.
2. A pillbox gives $D_{2n} - D_{1n} = \sigma_f$.
   Why: Its enclosed free charge is $\sigma_f A$.
3. A tiny electrostatic loop gives $E_{2t} = E_{1t}$.
   Why: Its circulation vanishes.
4. For linear isotropic media insert $D_i = \epsilon_i E_i$.
   Why: The constitutive law connects the auxiliary and physical fields.
5. If the fields are purely normal, $E_{2n} = (\sigma_f + \epsilon_1 E_{1n})/\epsilon_2$.
   Why: Solve the normal jump algebraically.

6. For $\sigma_f = 0$ and $\epsilon_2 = 3\epsilon_1$, $E_{2n} = E_{1n}/3$. Boundary conditions alone still need a source or far-field condition to fix its magnitude.
   Why: A relation between two unknown fields is not a complete solution.

**CHECK THE ENDPOINT**

Use normal D and tangential E conditions together with material laws and source/geometry information.

## T7 Linear dielectrics and capacitors

> Purpose: For an isotropic linear dielectric the polarization follows the macroscopic electric field.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

For an isotropic linear dielectric the polarization follows the macroscopic electric field.

1. Insert the linear P relation into $D = \epsilon_0 E + P$.
   Why: The material contribution is proportional to E.

2. Factor E to identify the permittivity.
   Why: Vacuum and material response add.

3. For a fully filled parallel-plate capacitor, $D = Q_f/A$, $V = Ed$, hence $C = \epsilon A/d$.
   Why: Symmetry makes the field nearly uniform away from edges.

**RESULT**

$\mathbf{P} = \epsilon_0\chi_e\mathbf{E}$, $\mathbf{D} = \epsilon\mathbf{E}$, $\epsilon = \epsilon_0(1+\chi_e) = \epsilon_0\epsilon_r$.

**EXAM TRAP**

State whether the battery remains connected. Fixed V and fixed Q lead to different energy changes.

**CHECK YOURSELF**

**If $\chi_e = 3$, what is the relative permittivity?**
Answer: $\epsilon_r = 4$.

This topic supports Q13, Q14, Q29, Q30.

## Work it through · Q13

Skill: Relative permittivity relation. Model · syllabus-based

**A linear dielectric has $\epsilon_r = 5$. Find its electric susceptibility.**

$\epsilon_r = 1 + \chi_e$, so $\chi_e = 4$. Both relative permittivity and susceptibility are dimensionless.

**CHECK THE ENDPOINT**

$\chi_e = 4$.

## Work it through · Q14

Skill: Fixed-voltage insertion. Model · syllabus-based

**A capacitor stays connected to an ideal battery while a dielectric completely fills it. How do capacitance and free charge change?**

For unchanged geometry, C increases by $\epsilon_r$. The battery fixes V, so $Q = CV$ also increases by $\epsilon_r$. Away from edges $E = V/d$ is unchanged.

**CHECK THE ENDPOINT**

Both C and Q multiply by $\epsilon_r$.

## Work it through · Q29

Skill: Capacitance calculation. Model · syllabus-based

**A fully filled parallel-plate capacitor has area $0.010$ m², separation 1 mm and $\epsilon_r = 4$. At 100 V find C, free Q and P.**

1. $C = \epsilon_0 \epsilon_r A/d = 3.542 \times 10^{-10}$ F.
   Why: Ignore fringe fields.
2. $Q_f = CV = 3.542 \times 10^{-8}$ C and $E = V/d = 10^5$ V/m.
   Why: The battery sets the potential difference.
3. $P = \epsilon_0(\epsilon_r - 1)E = 2.656 \times 10^{-6}$ C/m².
   Why: Use the linear constitutive law.

**CHECK THE ENDPOINT**

$C = 354.2$ pF; $Q_f = 35.42$ nC; $P = 2.656$ μC/m².

### Work it through · Q30

Skill: Isolated-capacitor energy. Model · syllabus-based

**A charged capacitor is disconnected from its battery before a dielectric of relative permittivity k is inserted fully. Derive the ratios of V and stored energy.**

1. Isolation fixes the free plate charge Q.
   Why: There is no route for charge exchange.
2. Capacitance becomes $kC_0$, so $V = Q/(kC_0) = V_0/k$.
   Why: Use the definition of capacitance.
3. $U = Q^2/(2C)$ therefore becomes $U_0/k$.
   Why: The energy formula must use the quantity held fixed.

**CHECK THE ENDPOINT**

$V/V_0 = 1/k$ and $U/U_0 = 1/k$; energy can be released as mechanical work.

## T8 Magnetic response of atoms

> Purpose: Orbital and spin magnetic moments explain different responses to applied fields.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

Orbital and spin magnetic moments explain different responses to applied fields.

1. A current loop is a magnetic dipole. Electrons also have intrinsic spin moments.
   Why: Both contribute to atomic magnetism.
2. An applied field changes orbital motion, inducing an opposing contribution.
   Why: This gives diamagnetism.
3. Existing moments tend to align, but thermal motion disrupts that order.
   Why: This gives weak paramagnetism in suitable materials.

**RESULT**

Diamagnets: $\chi_m < 0$; paramagnets: small $\chi_m > 0$; ferromagnets: strong, history-dependent response.

**EXAM TRAP**

Spin is an intrinsic quantum property, not literally a tiny charged sphere rotating classically.

**CHECK YOURSELF**

**Why does heating usually weaken paramagnetic alignment?**
Answer: More thermal disorder competes with field alignment.

This topic supports Q15, Q16.

## Work it through · Q15

Skill: Diamagnetic versus paramagnetic response.

**Legacy PYQ N25 Q9**

**Distinguish paramagnets and diamagnets.**

Diamagnets have a weak induced response opposing the applied field, with $\chi_m < 0$ and $\mu_r < 1$. Paramagnets have pre-existing moments that partially align with the field, giving small $\chi_m > 0$ and $\mu_r > 1$. Thermal disorder competes with paramagnetic alignment.

**CHECK THE ENDPOINT**

Diamagnets oppose weakly; paramagnets align weakly.

## Work it through · Q16

Skill: Paramagnetic temperature competition. Model · syllabus-based

**Why does a paramagnet have weak alignment rather than perfect alignment at ordinary temperature?**

Permanent atomic moments tend to align with an applied field, but thermal agitation disorders them. The resulting average alignment is small and normally decreases when temperature rises at fixed weak field.

**CHECK THE ENDPOINT**

Field alignment competes with thermal disorder.

## T9 Magnetization and magnetic torque

> Purpose: M measures magnetic dipole moment per volume, just as P measures electric dipole moment per volume.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

M measures magnetic dipole moment per volume, just as P measures electric dipole moment per volume.

1. Sum the atomic magnetic moments as vectors, then divide by volume.
   Why: Antialigned moments cancel.

2. For a small loop, $m = IA$.
   Why: Current and loop area determine its dipole moment.

3. In a uniform field its torque turns m toward B.
   Why: Opposite forces on opposite sides form a couple.

**RESULT**

$\mathbf{M} = \text{magnetic moment/volume}$ in A/m; $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$.

**EXAM TRAP**

M has units A/m, while magnetic moment m has units A m².

**CHECK YOURSELF**

**A moment $0.02$ A m² occupies $10^{-4}$ m³. What is M?**

Answer: $200$ A/m if uniformly aligned.

This topic supports Q31.

## Work it through · Q31

Skill: Magnetic dipole torque calculation.

**Prerequisite-course PYQ A26 Q10 · expanded**

**Obtain the torque expression for a planar current loop. Then consider a 100-turn coil of area $2 \times 10^{-3}$ m² carrying 0.20 A. Its normal makes 30° with B=0.50 T. Find moment, torque and dipole energy.**

1. For a rectangular loop, opposite magnetic forces of magnitude $IB\ell$ on a pair of sides form a couple. Their perpendicular separation gives $\tau = IAB\sin\theta$ for one turn, with θ the angle between its normal and B. Thus $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$, where $\mathbf{m} = IA\hat{n}$.
   Why: The Lorentz forces sum to zero in a uniform field but their moments need not cancel.

2. $m = NIA = 100(0.20)(0.002) = 0.040$ A m².
   Why: Moments of the turns add.

3. $\tau = mB\sin 30^\circ = 0.010$ N m.
   Why: Torque uses the angle of the normal, not the plane.

4. $U = -mB\cos 30^\circ = -0.0173$ J.
   Why: Choose zero dipole energy at a right angle.

**CHECK THE ENDPOINT**

$m = 0.040$ A m², $\tau = 0.010$ N m, $U = -0.0173$ J.

## T10 Bound currents and magnetized objects

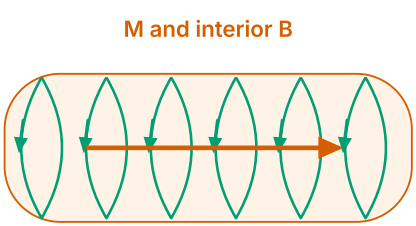
> Purpose: Tiny current loops cancel internally but leave equivalent currents at surfaces and where M varies.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

Tiny current loops cancel internally but leave equivalent currents at surfaces and where M varies.




**Surface loops act like a solenoid**

Long uniformly magnetized cylinder. The drawn loop sheet is the equivalent bound current, not a free winding.

1. Adjacent loops have opposing currents on their common edges.
   Why: Uniform bulk contributions cancel.
2. Loops at the outer edge have no neighbour to cancel them.
   Why: They form a surface current sheet.

3. A nonuniform loop density leaves a bulk remainder.
   Why: This is represented by curl M.

4. Use the ordinary magnetic field laws for these equivalent currents.
   Why: They reproduce the macroscopic field of the magnetization.

5. Define vector potential A by $B = \nabla \times A$. For localized steady sources, add $\mu_0 J_b \, d\tau/(4\pi\mathcal{R})$ and $\mu_0 K_b \, da/(4\pi\mathcal{R})$ over the body and surface to obtain A, where $\mathcal{R}$ is distance to the observation point.
   Why: Equivalent bound currents also act as sources for the magnetic vector potential.

**RESULT**

$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ in A/m; $\mathbf{J}_b = \nabla \times \mathbf{M}$ in A/m².

**EXAM TRAP**

The syllabus asks for physical interpretation of bound currents; the full microscopic derivation is not required.

**CHECK YOURSELF**

**For uniform M inside an object, what is the bulk bound current?**
Answer: Zero, although its surface current can remain.

This topic supports Q17, Q18, Q32, Q33.

## Work it through · Q17

Skill: Magnetic vector potential.

**Legacy PYQ N25 Q11 · expanded**

**Define magnetic vector potential. How can it describe the field of a magnetized object?**

The vector potential A is defined by $\mathbf{B} = \nabla \times \mathbf{A}$ and has units T m. In magnetostatics a localized magnetized object can be replaced by $\mathbf{J}_b = \nabla \times M$ and $\mathbf{K}_b = M \times \hat{n}$ in the usual current-source potential integrals. A is not unique: adding a gradient leaves its curl unchanged.

**CHECK THE ENDPOINT**

B is curl A; equivalent bound currents may be used as A sources.

## Work it through · Q18

Skill: Cancellation of microscopic loops.

**Legacy PYQ N25 Q18 · adapted marks**

**Discuss briefly the bound currents in a magnetized medium, emphasizing their physical interpretation.**

Microscopic magnetic dipoles can be modelled as current loops. Opposite currents on shared interior edges cancel for uniform M, leaving a surface current $K_b = M \times \hat{n}$. A spatially nonuniform magnetization can also leave $J_b = \nabla \times M$ in the bulk. These are equivalent macroscopic currents, not freely drifting conduction carriers.

**CHECK THE ENDPOINT**

Uncancelled microscopic loops produce bound surface and volume currents.

## Work it through · Q32

Skill: Uniformly magnetized cylinder. Model · syllabus-based

**A very long cylinder has uniform $\mathbf{M} = M\hat{\mathbf{z}}$ and no free current. Find its bound currents and B far from its ends.**

1. Uniform M gives $\mathbf{J}_b = 0$.
   Why: Its curl is zero.
2. On the curved surface, $\mathbf{K}_b = M\hat{\mathbf{z}} \times \hat{\mathbf{r}} = M\hat{\boldsymbol{\phi}}$; at flat ends the cross product is zero.
   Why: Use the outward surface normal.
3. The curved sheet is equivalent to a long solenoid with surface current M.
   Why: A solenoid's current loops have the same geometry.

**CHECK THE ENDPOINT**

$B_{\text{in}} = \mu_0 M$ axially and $B_{\text{out}} \simeq 0$, neglecting ends.

## Work it through · Q33

Skill: Nonuniform magnetization curl. Model · syllabus-based

**Inside a block, $\mathbf{M} = ax\hat{\mathbf{z}}$. Find its volume bound current and the surface current on the face x=L.**

1. Only $\partial M_z/\partial x = a$ is nonzero, so $\nabla \times \mathbf{M} = -a\hat{\mathbf{y}}$.
   Why: Use the Cartesian curl component $(\nabla \times M)_y = \partial_z M_x - \partial_x M_z$.

2. At x=L, the outward normal is $\hat{\mathbf{x}}$ and $\mathbf{K}_b = aL\hat{\mathbf{z}} \times \hat{\mathbf{x}} = aL\hat{\mathbf{y}}$.
   Why: The cross-product order is M then n.

**CHECK THE ENDPOINT**

$\mathbf{J}_b = -a\hat{\mathbf{y}}$; $\mathbf{K}_b(x = L) = aL\hat{\mathbf{y}}$.

## T11 The auxiliary field H

> Purpose: H removes the bound-current contribution from the magnetostatic circulation law.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

H removes the bound-current contribution from the magnetostatic circulation law.

1. Start with $\nabla \times \mathbf{B} = \mu_0(\mathbf{J}_f + \mathbf{J}_b)$.
   Why: Both kinds of steady current produce B.

2. Substitute $\mathbf{J}_b = \nabla \times \mathbf{M}$.
   Why: M represents the bound current.

3. Move curl M to the left and define H.
   Why: The remaining source is free current.

**RESULT**

$\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$; $\nabla \times \mathbf{H} = \mathbf{J}_f$; $\oint \mathbf{H} \cdot d\mathbf{l} = I_f$.

**EXAM TRAP**

H is not generally curl-free, and $H = 0$ does not imply $B = 0$ inside a magnet.

**CHECK YOURSELF**

**An ideal long solenoid has n turns per metre and free current I. What is H inside?**
Answer: $H = nI$ along the axis.

This topic supports Q19, Q34, Q39.

## Work it through · Q19

Skill: Definition of H. Model · syllabus-based

**Define H and state its magnetostatic circulation law.**

$\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$. Its circulation is $\oint \mathbf{H} \cdot d\mathbf{l} = I_{f,\text{enc}}$, using the right-hand orientation of the loop and current.

**CHECK THE ENDPOINT**

Only enclosed free current appears in the H circulation law.

## Work it through · Q34

Skill: Solenoid with magnetic filling.

**Prerequisite-course PYQ A26 Q16 · adapted to matter**

**A long solenoid has 1000 turns/m, current 0.50 A and a linear core with $\mu_r = 2$. Find H, M and B inside, and state the outside-field approximation. Recover the air-core case.**

1. Ampère's law for H gives $H = nI = 500$ A/m.
   Why: Only the free winding current enters.
2. $\chi_m = \mu_r - 1 = 1$, so $M = \chi_m H = 500$ A/m.
   Why: Use the specified linear response.
3. $B = \mu_0 \mu_r H = 1.257 \times 10^{-3}$ T.
   Why: B includes both free and bound-current effects.
4. For an ideal infinitely long solenoid the exterior field is zero; far from the ends of a long real one it is approximately zero. Setting $\mu_r = 1$ gives the air-core interior $B = \mu_0 nI$, while $H = nI$ and M=0.
   Why: The long-solenoid approximation neglects end fringing.

**CHECK THE ENDPOINT**

$H = 500$ A/m; $M = 500$ A/m; $B = 1.257$ mT.

## Work it through · Q39

Skill: Macroscopic magnetostatic description. Model · syllabus-based

**Explain how microscopic dipoles lead to M, bound currents and H. Derive the free-current Ampère law and discuss its limits.**

1. Define M as the vector magnetic moment per volume.
   Why: It averages many atomic moments.

2. Microscopic loop currents cancel on internal common edges but leave a surface sheet $K_b = M \times \hat{n}$; spatial variation leaves $J_b = \nabla \times M$.
   Why: These equivalent currents reproduce the macroscopic field.

3. Start with $\nabla \times B = \mu_0(J_f + J_b)$.
   Why: All steady currents source the magnetic curl.

4. Substitute the bound current and rearrange to $\nabla \times (B/\mu_0 - M) = J_f$.
   Why: Curl is a linear operation.

5. Define $H = B/\mu_0 - M$ and apply Stokes' theorem to obtain $\oint H \cdot dl = I_f$.
   Why: Surface curl and edge circulation are equivalent.

6. For linear isotropic matter, $M = \chi_m H$ and $B = \mu H$; in a ferromagnet this single constant law generally fails.
   Why: History and saturation change the response.

**CHECK THE ENDPOINT**

H isolates free current in the magnetostatic circulation law; it does not erase the material's contribution to B.

# T12 Magnetic susceptibility and permeability

Purpose: A linear isotropic magnetic medium has magnetization proportional to H.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

A linear isotropic magnetic medium has magnetization proportional to H.

1. Use $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$.
   Why: This is the definition of H rearranged.
2. Insert $\mathbf{M} = \chi_m \mathbf{H}$ and factor H.
   Why: Linearity makes a scalar permeability possible.
3. Apply this only over a range where the material response is approximately linear.
   Why: Ferromagnetic response can depend on both field and history.

**RESULT**

$\mathbf{M} = \chi_m \mathbf{H}$; $\mathbf{B} = \mu \mathbf{H}$; $\mu = \mu_0(1 + \chi_m)$.

**EXAM TRAP**

The electric relation uses P proportional to E; the magnetic convention uses M proportional to H.

**CHECK YOURSELF**

**If $\chi_m = -10^{-5}$, is $\mu$ above or below $\mu_0$?**

Answer: Slightly below.

This topic supports Q20, Q35.

## Work it through · Q20

Skill: Magnetic susceptibility definition.

**Legacy PYQ N25 Q3 · expanded**

**Define magnetic susceptibility and relate it to relative permeability for a linear isotropic material.**

Magnetic susceptibility is the dimensionless coefficient $\chi_m$ in $\mathbf{M} = \chi_m \mathbf{H}$. Since $B = \mu_0(H + M)$, $\mu_r = 1 + \chi_m$. For $\chi_m = 0.002$, $\mu_r = 1.002$. Ferromagnets need not obey one constant susceptibility.

**CHECK THE ENDPOINT**

Magnetic susceptibility measures magnetization per unit H in the linear regime.

### Work it through · Q35

Skill: Comparing electric and magnetic auxiliaries. Model · syllabus-based

**Compare P, D, M and H, and explain why $D = \epsilon_0 E$ and $B = \mu_0 H$ are not general material identities.**

1. P is electric moment density and M is magnetic moment density.
   Why: They describe material response.

2. $D = \epsilon_0 E + P$ and $B = \mu_0(H + M)$.
   Why: These definitions retain both vacuum and material terms.

3. Dropping P or M is justified only when that response vanishes or is being neglected.
   Why: A material can contribute substantial bound charge or current.

**CHECK THE ENDPOINT**

In linear matter use $D = \epsilon E$ and $B = \mu H$, with specified material assumptions.

## T13 Ferromagnetism and memory

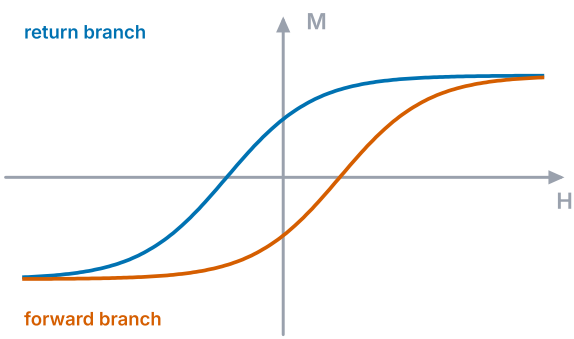
> Purpose: Domains can align cooperatively and retain alignment after the applied field is removed.

**BEFORE YOU START**

Use P1 for the physical quantities and M1 for the mathematics. A dipole is a separated pair of opposite charges; an electric current is moving charge. A vector has a magnitude and a direction.

**THE IDEA**

Domains can align cooperatively and retain alignment after the applied field is removed.




Illustrative M-H hysteresis branches, not measured material data. The zero-field intercept shows remanence; horizontal intercepts show coercive fields.

1. Within a domain many moments align.
   Why: Neighbouring moments interact cooperatively.
2. Changing an applied field moves domain walls and rotates moments.
   Why: Domains favourable to the field grow.

3. Removing the field does not retrace the initial path.
   Why: Pinning and collective rearrangement make the response history dependent.

4. Heating above the Curie temperature destroys spontaneous ferromagnetic order.
   Why: Thermal disorder overcomes the ordered state.

**RESULT**

Remanence is the remaining magnetization; coercive field is the reverse field needed to bring magnetization to zero on a hysteresis branch.

**EXAM TRAP**

A single constant susceptibility cannot describe an entire hysteresis loop. B-H and M-H coercive definitions need their own axes.

**CHECK YOURSELF**

**Why use a soft magnetic core in an AC device?**
Answer: Its narrow hysteresis loop reduces cycle losses.

This topic supports Q21, Q22, Q36, Q40.

## Work it through · Q21

Skill: Remanence and coercivity. Model · syllabus-based

**Explain remanence, coercivity and the role of domain behaviour in an M-H hysteresis loop.**

1. A ferromagnet contains domains with cooperative moment alignment.
   Why: Domains need not initially point in the same direction.
2. Applied H moves walls and rotates domains.
   Why: Favourable orientations grow.
3. After H returns to zero, pinning can leave nonzero M, called remanence.
   Why: The path depends on previous magnetic history.
4. A reverse H is needed to reduce M to zero; its magnitude on that branch is the coercivity.
   Why: The two intercepts describe different properties.
5. A narrow loop suits repeated cycling, whereas substantial coercivity helps a permanent magnet retain its state.
   Why: Applications depend on loss and resistance to demagnetization.

**CHECK THE ENDPOINT**

Remanence measures retained M; coercivity measures the required reverse H.

## Work it through · Q22

Skill: Curie transition. Model · syllabus-based

**Explain the effect of heating a ferromagnet through the Curie temperature, and distinguish the result from diamagnetism.**

1. Below the Curie temperature, interactions can maintain spontaneous alignment within domains.
   Why: Cooperative order survives thermal fluctuations.
2. Raising temperature increases disorder and reduces spontaneous magnetization.
   Why: Thermal agitation competes with ordering interactions.
3. Above the transition, spontaneous ferromagnetic order disappears and the material is paramagnetic.
   Why: It may still align weakly in an applied field.
4. This is not diamagnetism: a paramagnetic response is positive and comes from existing moments, whereas the diamagnetic contribution opposes the field.
   Why: The microscopic mechanisms differ.

**CHECK THE ENDPOINT**

Above its Curie temperature a ferromagnet loses spontaneous order and becomes paramagnetic.

## Work it through · Q36

Skill: Choice of magnetic material. Model · syllabus-based

**Explain why a transformer core and a permanent magnet require different hysteresis behaviour.**

1. A transformer cycles repeatedly, so low coercivity and a narrow loop reduce hysteresis loss.
   Why: Less irreversible rearrangement occurs each cycle.

2. High permeability allows useful flux with modest excitation.
   Why: The core responds strongly to the winding current.

3. A permanent magnet needs substantial remanence and high coercivity.
   Why: It must retain its state and resist unwanted demagnetization.

**CHECK THE ENDPOINT**

Soft narrow-loop material suits transformers; hard high-coercivity material suits permanent magnets.

## Work it through · Q40

Skill: Magnetic-material classification. Model · syllabus-based

**Compare diamagnetism, paramagnetism and ferromagnetism, including microscopic origin, field response and hysteresis.**

1. Diamagnetism is an induced response opposing the applied field, with small negative susceptibility.
   Why: Orbital motion changes under the field.

2. Paramagnetism aligns pre-existing moments weakly, giving small positive susceptibility.
   Why: Thermal disorder prevents complete alignment in weak fields.

3. Ferromagnetism has cooperative alignment within domains.
   Why: Interactions produce order that can persist without an applied field.

4. The initially weak net magnetization can grow by domain-wall motion and rotation.
   Why: Favourably oriented domains expand.

5. On cycling H, pinning produces hysteresis, remanence and coercivity.
   Why: The material does not follow a unique memory-free M(H) path.

6. Above the Curie temperature spontaneous order disappears; choosing a soft or hard material depends on whether repeated reversibility or retained magnetization is wanted.
   Why: Temperature and history are essential parts of the material description.

**CHECK THE ENDPOINT**

Diamagnets oppose weakly, paramagnets align weakly, and ferromagnets show cooperative, often hysteretic response.

# Formula and condition sheet

## T1 · Dielectrics and induced dipoles

$\mathbf{p} = \alpha \mathbf{E}_{\text{local}}$ in the weak-field isotropic model.

## T2 · Permanent dipoles: torque and force

$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$, $U = -\mathbf{p} \cdot \mathbf{E}$; for a fixed dipole, $\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}$.

## T3 · Polarization as a density

$\mathbf{P} = \text{dipole moment}/\text{volume}$; units C/m².

## T4 · Bound charge and the field of polarization

$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$; $\rho_b = -\nabla \cdot \mathbf{P}$.

## T5 · Displacement and free charge

$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$; $\nabla \cdot \mathbf{D} = \rho_f$; $\oint \mathbf{D} \cdot d\mathbf{a} = Q_f$.

## T6 · Electric boundary conditions

$D_{2n} - D_{1n} = \sigma_f$; $\mathbf{E}_{2t} = \mathbf{E}_{1t}$ in electrostatics. Normal n points from medium 1 to 2.

## T7 · Linear dielectrics and capacitors

$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, $\mathbf{D} = \epsilon \mathbf{E}$, $\epsilon = \epsilon_0(1 + \chi_e) = \epsilon_0 \epsilon_r$.

## T8 · Magnetic response of atoms

Diamagnets: $\chi_m < 0$; paramagnets: small $\chi_m > 0$; ferromagnets: strong, history-dependent response.

### T9 · Magnetization and magnetic torque

$\mathbf{M} = \text{magnetic moment/volume}$ in A/m; $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$.

### T10 · Bound currents and magnetized objects

$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ in A/m; $\mathbf{J}_b = \nabla \times \mathbf{M}$ in A/m².

### T11 · The auxiliary field H

$\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$; $\nabla \times \mathbf{H} = \mathbf{J}_f$; $\oint \mathbf{H} \cdot d\mathbf{l} = I_f$.

### T12 · Magnetic susceptibility and permeability

$\mathbf{M} = \chi_m \mathbf{H}$; $\mathbf{B} = \mu \mathbf{H}$; $\mu = \mu_0(1 + \chi_m)$.

### T13 · Ferromagnetism and memory

Remanence is the remaining magnetization; coercive field is the reverse field needed to bring magnetization to zero on a hysteresis branch.

## Past-paper route

**Q2 → T1**
Legacy PYQ N25 Q5 · expanded

**Q8 → T4**
Legacy PYQ N25 Q10 · expanded

**Q15 → T8**
Legacy PYQ N25 Q9

**Q17 → T10**
Legacy PYQ N25 Q11 · expanded

**Q18 → T10**
Legacy PYQ N25 Q18 · adapted marks

**Q20 → T12**
Legacy PYQ N25 Q3 · expanded

**Q27 → T5**
Legacy PYQ N25 Q19 · expanded

**Q31 → T9**
Prerequisite-course PYQ A26 Q10 · expanded

**Q34 → T11**
Prerequisite-course PYQ A26 Q16 · adapted to matter

**Q37 → T4**
Legacy PYQ N25 Q17 · adapted marks